



Before: Two connected Z spiders

After: Single fused Z spider

Figure 1: The Spider Theorem in ZX calculus: two connected Z spiders fuse into a single Z spider. This fundamental rule enables simplification of ZX diagrams and reasoning about quantum operations. Spider Theorem (Fusion): When two spiders of the same color are connected, they fuse into a single spider. The total number of wires is preserved: $(m, n) \cup (n, p) = (m, p)$.